CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 15 : Python Program to Implement Decision Tree Classifier

Aim

To develop a Python program that classifies a given number using a Decision Tree classifier from scikit-learn.

Algorithm

- Import DecisionTreeClassifier from sklearn.tree.
- Define the training data x (numbers 1 to 5) and the class labels y (0, 0, 0, 1, 1).
- Create the Decision Tree model and train it using fit().
- Read a number from the user.
- Predict the class of the number using predict().
- Print "Yes" if the predicted class is 1; otherwise print "No".

Program

```
from sklearn.tree import DecisionTreeClassifier
x = [[1], [2], [3], [4], [5]]
y = [0, 0, 0, 1, 1]
model = DecisionTreeClassifier()
model.fit(x, y)
n = int(input("Enter a number: "))
result = model.predict([[n]])
if result[0] == 1:
    print("Yes")
else:
    print("No")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/shaik/OneDrive/Documents/15.py ===============
Enter a number: 4
Yes
>>>
```

RESULT

Thus, the Decision Tree classifier was implemented successfully using Python.